

A discrete macro-element method for structural assessment of masonry arch bridges

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ABSTRACT

Masonry Arch Bridges (MABs) are complex structures whose behaviour is dependent on the interaction of various components. Over the years, a variety of modelling strategies have been proposed to capture this behaviour, each offering a different balance between computational cost, accuracy, and level of detail. This paper investigates the effectiveness and accuracy of a Discrete Macro-Element Modelling (DMEM) strategy for the analysis of MABs. Originally developed for the analysis of unreinforced masonry buildings (URM), the DMEM strategy is here extended to describe the behaviour of masonry and non-masonry bridge components (including the backfill) as well as their non-linear interactions under the action of external vertical loads. To enable this application, a novel three-dimensional hexahedral macro-element has been developed and implemented, allowing interaction with adjacent elements through all the six macro-element's faces.

An extensive numerical investigation is carried out to validate the developed strategy against several experimental and numerical benchmarks, characterised by different bridge typologies, geometries and loading conditions. This includes single- and multi-span bridges, with attached and detached spandrel walls and arch brickwork defects such as ring separation. The results show that the DMEM approach can accurately predict the load-carrying capacity of MABs while maintaining a very low computational cost, making it suitable for the assessment of larger and complex structures by both engineers and researchers.

1. Introduction

Masonry Arch Bridges (MABs) are long-lasting structures that continue to contribute significantly to the functioning of transportation infrastructure in many countries [1,2]. Renowned for their robustness, durability, and aesthetic value, MABs have been fundamental in developing transportation and trade networks across diverse countries. Beyond their functional utility, masonry arch bridges hold significant cultural and social value, making their preservation essential to safeguard architectural and historical heritage.

The assessment of the structural integrity of MABs presents considerable challenges, one of the most critical being the accurate assessment of their capacity against traffic loadings [3–5] and natural events like

scouring [6–8] that may jeopardize their functioning and stability. Masonry is characterized by its discontinuous nature, exhibiting non-linear, anisotropic behaviour that is highly influenced by the type of masonry bond and the loading direction. Mortar joints often emerge as the most vulnerable component within this structure [9]. These unique characteristics must be well understood to create accurate mechanical models of all MAB components (e.g. arches, piers, spandrel walls). The role of the backfill in the overall MAB response should also be explicitly accounted for due to its significant influence (see e.g. [10,11]).

Over the past decades, extensive experimental and numerical research has been undertaken to capture the complex 3D behaviour of MABs (see [3] for a review of these, and [12] for recent experimental tests). This progress has greatly advanced the understanding of MAB

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mechanics and the accuracy of numerical strategies for predicting their behaviour under service loading and at collapse. A wide range of modelling approaches is now available for assessing MABs [13], spanning semi-empirical and graphical methods (see e.g. [14], limit state analysis-based approaches focused on collapse behaviour [15–18], and advanced numerical techniques capable of describing performance under both service and ultimate loads; the latter are briefly summarized below.

The Finite Element Method (FEM) has progressively advanced in the analysis of masonry arch bridges (MABs), evolving from early 1D nonlinear beam models (see e.g. [19]) to more sophisticated 3D frameworks (e.g. [6,20,21]). More advanced 3D modelling strategy based on FEM have also been developed, in which masonry components are represented at the mesoscale, explicitly accounting for masonry–backfill interaction (see e.g. [7,22,23]).

The Discrete Element Method (DEM) models a masonry arch bridge as an assembly of rigid blocks separated by zero-thickness joints [24]. Each block can translate, rotate, and detach, while the solver continually detects and updates contacts. As a result, large displacements, sliding, and opening are captured without mesh distortion. This discontinuum description can closely mirrors the actual brick-and-mortar fabric of masonry structures. Consequently, DEM has been applied in three dimensions to simulate entire bridges [24–28], including arches, spandrel walls, and backfill. These models have yielded realistic estimates of load-carrying capacity, crack propagation, and collapse mechanisms, while also clarifying how backfill properties and soil–structure interaction influence overall behaviour.

Since its conception [29], the Discrete Macro-Element Method (DMEM) has emerged as a promising and innovative approach for the analysis of masonry structures, complementing the more established FEM and DEM. The DMEM is based on shear deformable elements with essential degrees of freedom that interact through nonlinear discrete interfaces in which the membrane behaviour is embedded. Its development was driven by the need to accurately represent the behaviour of masonry components while ensuring computational efficiency, making it particularly well-suited for the structural assessment and preservation of URM historic structures. As such, the DMEM occupies a distinctive position among existing modelling strategies, combining unique capabilities. Unlike the DEM, DMEM retains discrete interactions between adjacent components but allows element deformation to enhance kinematic representation also at the macro-scale. Compared to nonlinear FEM simulations, DMEM offers a computational advantage due to its reduced degrees of freedom, albeit with certain simplifications.

While DMEM has been successfully applied and validated across numerous masonry structures [29–32], its application to MABs has been relatively limited. Initial validations of the DMEM for bridge modelling involved comparisons with 3D non-linear FEM simulations of a railway bridge located in Italy [33,34] and a medieval masonry bridge located in Portugal [35]. Another study further investigated the capabilities of DMEM for evaluating the impact of pier settlements on the performance of a masonry arch bridge [36]. However, this analysis did not include validation against experimental data or other accurate numerical analyses. Thus, a more in-depth investigation involving comparisons with experimental test results is required to validate the DMEM and explore its capabilities, limitations and future developments. These investigations should also shed light on the possibility of using the DMEM approach for describing the behaviour of the backfill material, and on its effectiveness in simulating the key contribution of soil to the behaviour of MABs (i.e. to distribute loads and provide weight and passive resistance [10,37]).

This article aims to address the research needs described above by presenting the results of an extensive numerical campaign examining the effectiveness of the DMEM for evaluating the performance and load-carrying capacity of MABs. For this purpose, the results of the tests conducted at the Bolton Institute are considered [38,39]. In these tests, large-scale single- and multi-span masonry arch bridges with multi-ring

arches have been subjected to increasing vertical loads at deck level up to collapse. Tests have also been performed on bridges with attached and detached spandrel walls, and with arch brickwork defects such as ring separation.

The DMEM adopted in this study has been integrated into the software HiStrA (Historical Structures Analysis), which includes a pre- and post-processor specifically developed for masonry arch bridges (HiStrA-Bridge module) [40]. To enhance the robustness of the obtained results, the DMEM model results were also compared with those obtained from other numerical models [21–23]. The results of this study aim to unlock the adoption of HiStrA-Bridge by practitioners and researchers interested in the assessment and preservation of historic MABs.

2. DMEM strategy for MABs

This section illustrates the application of the Discrete Macro-Element Modelling (DMEM) strategy [32] to characterize the non-linear behaviour of MABs. According to this strategy, the MAB components (including the backfill) are discretised into macro-portions, whose behaviour is simulated by corresponding equivalent mechanical elements. Differently from the classical DEM approach [25,26], in which each element is considered as a rigid body, the adopted macro-elements possess shear deformability. The mechanical interaction among adjacent macro-elements is described by zero-thickness interfaces with uniformly distributed non-linear links incorporating the non-linear behaviour of the macro-portion, following a fibre discretisation strategy. In the DMEM, axial and flexural behaviours are intentionally concentrated at the interfaces, in contrast to the FEM, where such effects are distributed along the element. As a result, allowing penetration between adjacent elements is essential to accurately capture these behaviours.

The macro-element can describe the main failure mechanisms of masonry components, namely flexural failure, shear-diagonal failure, shear-sliding failure, and out-of-plane flexural and torsional failure. However, the computational cost of the proposed numerical approach is greatly reduced compared to traditional non-linear FEM and DEM simulations without significantly sacrificing the accuracy of the results. Unlike FEM, which often simplifies masonry structures into a homogenised medium, DMEM recognises the discrete nature of these constructions. DEM, while detailed, becomes impractical for large-scale applications due to its high computational burden.

2.1. 3D macro-element mechanical parameters

The DMEM macro-element was originally developed to simulate the in-plane non-linear response of masonry walls subjected to in-plane loading [30]. A simplified mechanical scheme was introduced to model a masonry wall portion, consisting of an articulated quadrilateral element formed by four rigid edges connected by hinges and a diagonal non-linear spring. This basic element (Fig. 1a), referred to as a panel, can simulate the main collapse mechanisms of masonry walls under horizontal and vertical loads within its plane, including flexural failure, diagonal shear failure, and sliding shear failure. The interaction between adjacent panels is governed by zero-thickness interfaces corresponding to the shared lateral surfaces. The 2D formulation of the macro-element involves four degrees of freedom: three associated with in-plane rigid-body motion and one for in-plane shear deformability. This configuration allows for an accurate representation of the wall's kinematics and failure mechanisms through proper calibration of the non-linear links.

In terms of kinematics, the diagonal link depends on the in-plane shear degree of freedom of the corresponding panel. Conversely, the interface kinematics are governed by a linear combination of the degrees of freedom of the adjacent elements [32]. The original model was extended to describe the three-dimensional behaviour of masonry by introducing additional non-linear links and degrees of freedom [31]. Specifically, the seven degrees of freedom of the enhanced model allow for the description of both in-plane and out-of-plane rigid body motions,

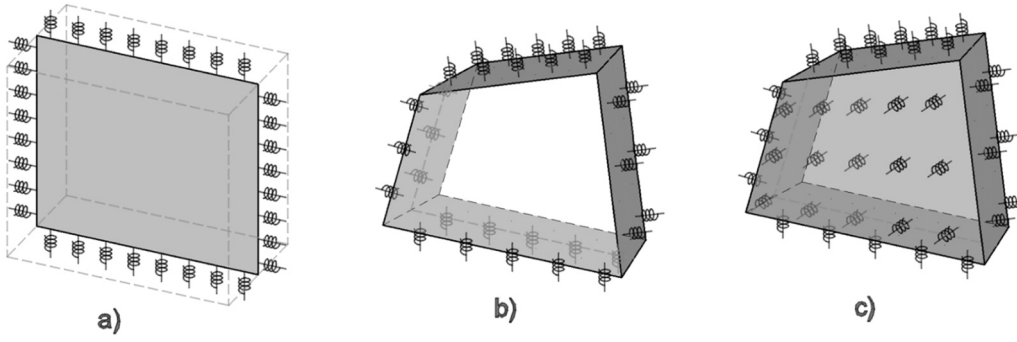


Fig. 1. Evolution of the DMEM approach: (a) plane regular element, (b) irregular spatial element and (c) irregular spatial element with interfaces on all the faces [33,42].

as well as in-plane shear deformability. The 3D interfaces in this model are equipped with longitudinal and additional transversal sliding springs, which control both in-plane and out-of-plane sliding mechanisms and torsion around the axis perpendicular to the plane of the interface. Subsequently, a more versatile shell macro-element was introduced, designed for discretizing structures with curved geometries [41]. Fig. 1b illustrates the irregular spatial element and its surrounding interfaces, highlighting the model's capability to handle complex geometries. The element exhibits an irregular geometry and can accommodate skew interfaces with respect to the plane of the element with variable thickness.

In this study, in order to improve the description of the three-dimensional behaviour of masonry arch bridges, the previously developed element of [41] is enhanced by introducing interaction surfaces between adjacent elements on all the six faces rather than only four faces (Fig. 1c).

Fig. 2a illustrates the kinematics of the element, showing the degrees of freedom $u_x, u_y, u_z, \phi_x, \phi_y, \phi_z, \gamma$ of two connected elements I and J . The first six correspond to the rigid body motion (dash-dot blue line configuration), while the final term represents in-plane shear deformability. Additionally, the same figure highlights how the displacements $\underline{u}_I$ and $\underline{u}_J$ along the interface between two adjacent elements depend on their respective degrees of freedom.

One of the crucial aspects of the DMEM framework is the mechanical calibration of the nonlinear components of the element, i.e., the links of the interfaces and the internal diagonal link. Specific assumptions regarding material properties, structural geometry, and element interactions are necessary to the purpose. Each macro-element represents a finite portion of the masonry wall, generally treated as a homogenized material. The zero-thickness interfaces manage the membrane-flexural response and the shear-sliding behaviour of adjacent elements, while

in-plane shear deformability is linked to the angular distortion of the articulated quadrilateral. The mechanical characterization of these zero-thickness interfaces is conducted through a straightforward fibre calibration process (Fig. 2b). The in-plane shear element deformability is calibrated to match a reference geometric-consistent continuous model by means of a diagonal shear link, Fig. 3. The non-linear links within the interface are categorized into orthogonal and shear-sliding types (Fig. 4).

2.1.1. Orthogonal links

Orthogonal links are employed to replicate membrane and flexural deformations of the panel along specific directions. They concentrate these deformations at the zero-thickness interfaces where the panels connect, thereby accurately capturing the directional behaviour of masonry components.

The orthogonal non-linear links model the mechanical

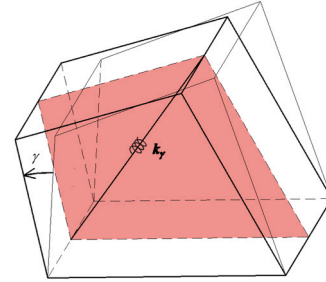


Fig. 3. Shear element deformability introduced by means of the diagonal shear link with elastic stiffness k_r .

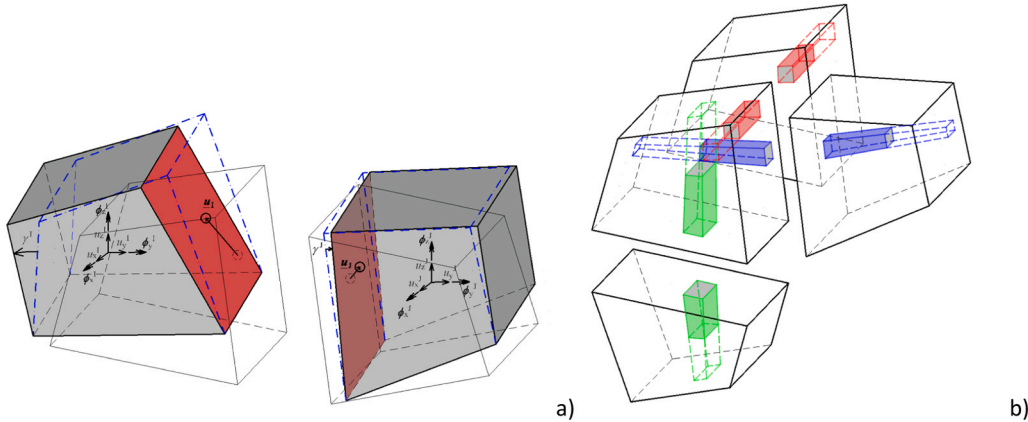


Fig. 2. (a) Kinematics of the element with interface between two contiguous elements (in red) in deformed configuration and (b) fiber calibration approach for transversal links interface.

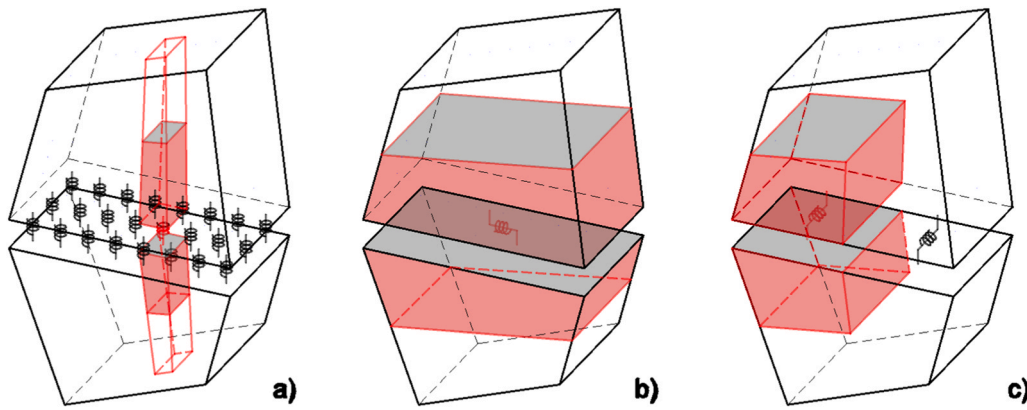


Fig. 4. Interface's links calibration: (a) Orthogonal N-Links; (b) In-plane sliding N-links; (c) Out-of-Plane sliding N-Links [42].

characteristics of the element, treating the masonry as an orthotropic homogeneous medium. These links describe the non-linear behaviour of the corresponding fibre along specific material directions. The calibration of each link is based on the assumption that the masonry strip represents a homogeneous inelastic material. This calibration may also account for cyclic behaviour governed by the fracture energy values for tensile and compressive responses. Additionally, these links adhere to various laws governing the post-elastic behaviour (e.g., linear hardening, linear softening, exponential softening, and parabolic).

In line with the adopted fibre calibration strategy, the flexural stiffness of the panel is calibrated by assigning to each link the axial stiffness and strength corresponding to the masonry strip it represents, as shown in Fig. 4a. The dimensions and influence area of each masonry strip are defined by half the panel dimension in the direction perpendicular to the interface. Since in the present study interfaces have been introduced along the top and bottom faces of the element, in addition to those already present along the four lateral faces, the fibre calibration procedure for the relevant orthogonal links has been extended to account for a third direction, as shown in Fig. 2b.

2.1.2. Shear-sliding links

Non-linear shear-sliding links along the interface govern the torsional and shear-sliding behaviour. A single link is utilized for the in-plane model, whereas three non-linear links are employed in the spatial models to capture masonry's potential failure modes adequately. This set of links is the minimal configuration necessary to replicate all plausible failure scenarios in masonry structures [31].

One spring describes the main in-plane shear-sliding behaviour, which enables the element's sliding along the interface and is calibrated using a rigid-plastic Mohr-Coulomb law (Fig. 4). Two parallel springs describe the out-of-plane shear deformability. These springs describe the out-of-plane sliding behaviour and the connected adjacent panels' elastic and inelastic torsional responses around the axis perpendicular to the interface plane. The out-of-plane shear deformability of each link connecting two adjacent panels is calibrated based on their influence volumes (Fig. 4c). Furthermore, the yielding strength of each link is determined using the Mohr Coulomb criterion, by considering the current interface contact area and axial force level (corresponding to the links orthogonal to the interface). For the interfaces located on the top and bottom faces of the element, the full shear deformability is concentrated in the sliding links.

2.1.3. Diagonal link

The diagonal non-linear shear behaviour of a masonry panel is described by a single degree of freedom, directly related to the elongation of a diagonal non-linear link (Fig. 3). With regards to the elastic phase, the stiffness k_r of the diagonal shear link is calibrated by establishing an energy equivalence between the articulated quadrilateral,

governed by the diagonal spring, and a continuous reference elastic model. The yielding force is determined based on the reference model's tensile or compressive stress limits, with post-elastic behaviour dictated by a chosen constitutive law. Various yielding criteria, which are highly influenced by the compressive stresses within the element, can be utilized to evaluate the shear capacity. The Mohr-Coulomb or Turnsek-Cacovic laws are typically used, although alternative constitutive laws may also be applied. For further details about the calibration procedure, reference can be made to [32,43].

3. Modelling of case studies

The adopted DMEM strategy for the analysis of MABs is validated against the results of a comprehensive experimental campaign carried out at the Bolton Institute in the United Kingdom over the course of the 1990s [38,39]. This campaign encompassed a series of tests on large-scale multi-ring brickwork MAB prototypes, consisting of single and multiple spans with and without attached spandrel walls and various arch ring defects. The results of this extensive campaign have been paramount for the calibration and validation of several numerical modelling strategies developed over the years (e.g. [21–23]), by ensuring that the complex behaviours of MABs is accurately simulated under a wide range of conditions.

In addition to the experimental campaign outcomes, comparisons are made with results from other numerical modelling strategies, including continuum macro-scale masonry discretisation methods [21] and mesoscale approaches [22,23]. These comparisons serve to validate the DMEM models, particularly when notable differences arise between numerical simulations and experimental results. This approach highlights both consistencies and discrepancies across different numerical methodologies, offering a comprehensive evaluation of the accuracy and robustness of DMEM models.

3.1. Bridge specimens and experimental test procedures

The following subsections illustrate the various analyses conducted to investigate the capability of DMEM to model MABs while accounting for the interaction between different bridge elements. The analyses encompass various bridge specimens designed to explore specific structural characteristics and defects commonly found in MABs.

Six single-span bridge prototypes tested in Salford are numerically investigated. These include two 3 m-span MABs with detached spandrel walls (*Bridge 3.1* and *Bridge 3.2*), with the latter also featuring ring separation. Two 5 m-span MABs are also studied: *Bridge 5.1* with detached spandrel walls, and *Bridge 5.2*, a variant incorporating defects at the interfaces between arch rings. The study also investigates numerically a defect-free 3 m span MAB with attached spandrel walls (*Bridge 3.3*) and another 3 m span MAB with attached spandrel walls but weak

connections between rings (Bridge 3.4).

In addition to the single-span prototypes, the study includes the simulation of the experimental tests on three large-scale model bridges, each comprising three 3 m spans. These multi-span bridges were designed to mirror the characteristics of their single-span counterparts. *Bridge MS1* and *Bridge MS3* featured attached spandrel walls but were subjected to different loading conditions, while *Bridge MS2* incorporated detached spandrel walls across all spans. Table 1 reports the geometric characteristics and details of defects of the bridge specimens.

All 3 m-span bridges specimens were constructed with two-ring arch barrels using the stretcher bond method in a segmental circular configuration. Each arch has a thickness of 215 mm. Fig. 5 illustrates the detailed geometric features of two representative specimens: *Bridge 3.1* (single-span) and *Bridge MS2* (multi-span).

A vertical transverse line load was applied in the tests through a rigid steel beam positioned directly on the surface of the backfill and supported by a reaction frame. The line load was distributed uniformly across almost the arch width and increased monotonically until failure. This procedure allowed for a thorough analysis of the damage patterns, arches' structural integrity and load-bearing capacity under controlled conditions.

Throughout the testing program, extensive instrumentation was employed to gather detailed data on the bridges' performance. The setup included displacement transducers, surface-mounted strain gauges, and soil pressure cells strategically placed in the backfill and at the soil-structure interfaces. The instrumentation was specifically chosen and positioned to achieve two primary objectives: (a) to accurately monitor the levels of applied loads and the corresponding global structural displacements; (b) to observe the mobilization of backfill pressures. Reference can be made to [38,39] for additional information and a more comprehensive description of the prototypes, setup, and results of the tests.

3.2. Materials

The experimental campaign encompassed an extensive series of material and component tests to accurately characterize the construction's mechanical material's properties. The brickwork comprised full-size Class A engineering bricks bonded with a mortar mix ratio of 1:2:9 (cement:lime:sand). Compressive tests were conducted to determine its compressive strength and the initial tangent modulus of elasticity. Additionally, shear tests were performed on several small brickwork panels to evaluate the shear bond strength and the coefficient of friction for the mortar joints. To determine the unit weight of the brickwork, measurements were taken of the individual unit weights of the bricks and the mortar, which were then combined in their respective proportions. A large shear box apparatus was employed to characterise the backfill material, providing data on cohesion and the friction angle. Moreover, in-situ density measurements were conducted to obtain the bulk unit weight of the backfill.

It is noteworthy that the test results provided information on most of the parameters required for the DMEM model. For brickwork, the available data include specific weight (γ_m), initial tangent modulus (E_m), compressive strength (f_{cm}), sliding-shear strength (c_m), and the coefficient of friction of mortar joints (Φ_m). For cases of defects at the interfaces between arch rings (ring separation), [38] specifies that this interface, particularly for brickwork specimens constructed with 'damp sand,' has a friction coefficient of 0.53. In the case of backfill, the data encompass weight (γ_b), friction angle (Φ_b), and cohesion (c_b).

There are notable gaps in the experimental data, specifically there is no information on Poisson's ratio (ν) tensile strength (f_t) and fracture energies in compression and tension (G_c , G_t) of masonry, and on the elastic modulus (E_b) of the backfill material. These deficiencies were addressed by integrating supplementary data from other numerical validations [21–23]. The collated material properties have been consolidated and are presented in Table 2. More specifically, in Zhang et al. [22], the tensile strength of the masonry (f_{tm}) was evaluated at 0.2 MPa, and the tensile fracture energy was evaluated at 0.02 N/mm (G_{tm}). For the backfill, the Young's modulus was assumed equal to 200 MPa (E_b), a Poisson's ratio of 0.20 (ν_b), a nominal cohesion of 0.001 MPa (c_b), and a friction angle of 55° (ϕ_b). This friction angle contrasts with the 60° reported by [38], which [21] noted is slightly higher than typical values observed for cohesionless materials. A notable point was the assumption of a non-zero cohesion value for backfill by [22], albeit close to zero. This assumption was made to address convergence issues in the modelling process. Meanwhile, Tubaldi et al. [23] aligned with the values chosen by [22], but diverged in specifying the tensile fracture energy at 0.05 N/mm (G_{tm}) and introduced a compressive fracture energy of 1 N/mm (G_{cm}). Zizi et al. [21] employed an iterative calibration process to deduce missing data, resulting in a Young's modulus of 50 MPa for the backfill (E_b), a tensile strength of 0.2 MPa for the masonry (f_{tm}) and a friction angle of 30° at the interface between the masonry and backfill (ϕ_{bm}). Furthermore, the value of 0.15 for masonry Poisson's ratio ν_m was established based on [44].

It is noteworthy that the current version of the DMEM-element does not account for Poisson effect, i.e., compression along one direction does not result in expansion or contraction in the perpendicular direction. Moreover, as detailed in Section 2, the definition of the DMEM-element properties follows a fibre calibration procedure, requiring compression and tension resistance parameters for effective modelling. However, soil typically does not exhibit a distinct compression resistance.

Considering that the soil's primary role in MABs is to distribute applied loads and provide counterbalancing weight and passive resistance, the lack of Poisson's effect and the difficulty in defining the soil compression resistance should not play a crucial role in the bridge behaviour. Nevertheless, a suitable compressive strength value must be assigned to prevent local failures of the backfill due to excessive soil compression and ensure that soil does not artificially contribute to the load-bearing capacity of the bridge, maintaining the accuracy of the simulation. In the models, a compressive strength value of 0.9 MPa was

Table 1
Geometric characteristics and details of defects of the bridge specimens [38,39].

Bridge	Span-length (m)	Width (m)	Number of rings	Defects	Experimental failure load (kN)	Load configuration
3.1	3	2.88	2	spandrels detached	540	load at the quarter span
3.2	3	2.88	2	spandrels detached	360	load at the quarter span
				ring separation		
3.3	3	3.54	2	-	600	load at the quarter span
3.4	3	3.54	2	ring separation	320	load at the quarter span
5.1	5	3.01	4	spandrels detached	1720	load at the quarter span
5.2	5	3.01	4	spandrels detached	500	load at the quarter span
				ring separation		
MS1	3 × 3	3.54	2	-	455	load at the quarter span of the central arch
MS2	3 × 3	2.88	2	spandrels detached	320	load at the quarter span of the central arch
MS3	3 × 3	2.54	2	-	325	load at 240 mm from the north of the crown of the central arch

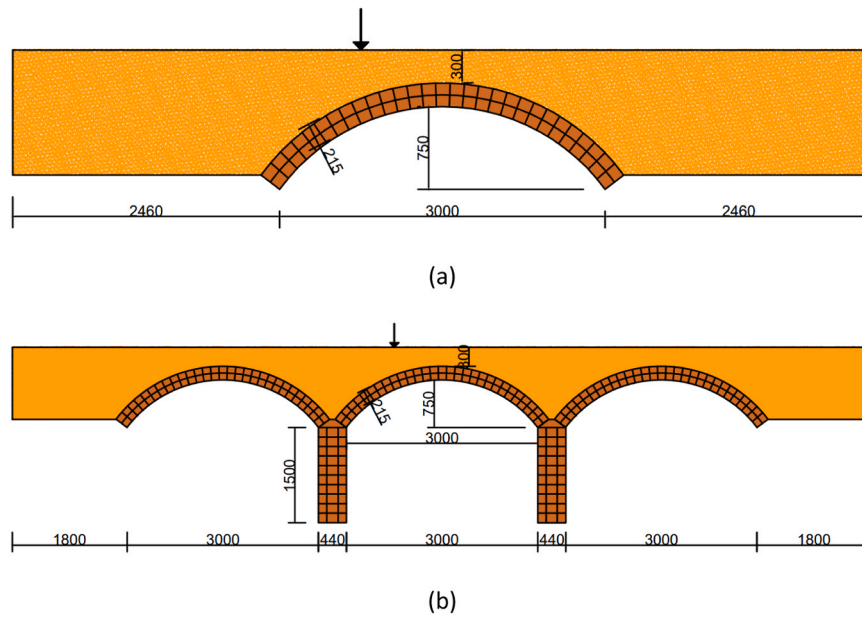


Fig. 5. Geometric characteristics of the bridge specimens (in millimetres): (a) Bridge 3.1 (adapted from [38]); (b) Bridge MS2 (adapted from [39]).

Table 2
Material properties.

Parameters with constant values throughout various tests							
Material	γ (kN/m ³)	ν (-)	Φ (-)	c (MPa)	f_t (MPa)	G_t (kN/m)	G_c (kN/m)
Masonry	-	0.15	-	0.29	0.2	0.02	3
Backfill	22.2	0.2	1.47	0.001	0.001	0.01	-
Parameters with different values throughout various tests							
Bridge	γ_m (kN/m ³)	E_m (GPa)	f_{cm} (MPa)	Φ_m (-)	E_b (GPa)		
3.1	22.7	15.9	26.5	0.64	0.2		
3.2	22.7	15.9	26.5	0.64	0.2		
3.3	22.7	15.9	26.5	0.5	0.2		
3.4	22.7	15.9	26.5	0.5	0.2		
5.1	22.7	9.2	18.1	0.64	0.2		
5.2	22.7	25.0	28.2	0.64	0.2		
MS1	22.4	16.2	26.8	0.5	0.05		
MS2	22.4	16.2	26.8	0.5	0.05		
MS3	22.4	22.3	27.7	0.5	0.05		

assigned to the soil. This value was chosen to ensure that the soil's behaviour aligns with the simulation's objectives, particularly in preventing local backfill failures and preserving realistic load distribution. It is also worth noting that the numerical simulation of the backfill behaviour still represents an open issue in the scientific community, and that the most advanced models used to date are Mohr Coulomb/Drucker Prager models (see e.g. [22]).

3.3. Numerical modelling strategy

The proposed modelling strategy is embedded within HiStrA Bridges software, Version 2024.1.1 [40]. The software provides a user-friendly graphical interface designed to aid in modelling masonry arch bridges. The models implemented in this study are generated through parametric input, although the software also supports an alternative strategy for geometry definition via CAD file import routines. The software's interface streamlines the specification of geometric data, material properties, and loading conditions, thereby enhancing the efficiency of the model setup process.

For bridges with detached spandrel walls, the experimental specimens were constructed with a narrow gap (approximately 10–15 mm wide) between the inner face of the spandrel wall and the edge of the

arch barrel. This detail ensured that the spandrel walls were not mechanically connected to the arch barrel, while still providing lateral confinement to the backfill. In the numerical analyses of these specimens, a strip model is employed (Fig. 6), featuring a singular set of elements across the width of the arch and backfill. This approach, aimed at computational efficiency, yields precise outcomes for square arch bridges under uniformly distributed line loads [21–23].

In contrast, the analysis of bridges with attached spandrel walls requires a three-dimensional model with multiple macro-elements along the transverse direction (Fig. 7). At a minimum, a model with very low computational cost employs two lines of elements for the spandrel wall, two lines between the spandrel wall and the edges of the load print, and one for the area where the load is applied. However, in the final model, two elements are used for the area where the load is applied, ensuring a more accurate representation.

The approach for evaluating bridges with defective rings aligns with the previously mentioned strategies, incorporating a critical adjustment through the inclusion of weak interfaces between arch rings (Fig. 8). Specifically, ring separation in numerical analyses is modelled by friction interfaces that represent the physical defects, capturing the independent behaviour of individual rings.

In the DMEM formulation, the constitutive behaviour associated with each failure mechanism is governed by distinct material laws tailored to capture the non-linear mechanical response of masonry. For flexural failure (also referred to as the rocking mechanism), a linear softening law is adopted for both compressive and tensile responses, calibrated based on the corresponding fracture energy. This allows for a realistic representation of progressive damage and energy dissipation during crack propagation. The diagonal cracking or shear failure

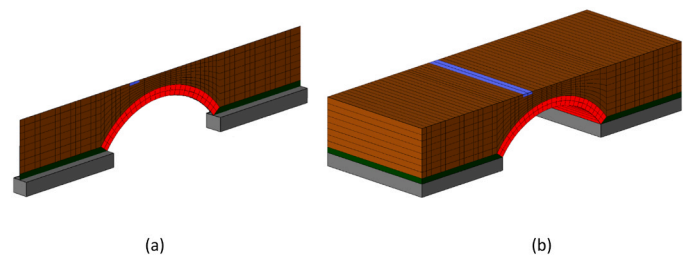


Fig. 6. Strip model: (a) Numerical model; (b) Geometric model.

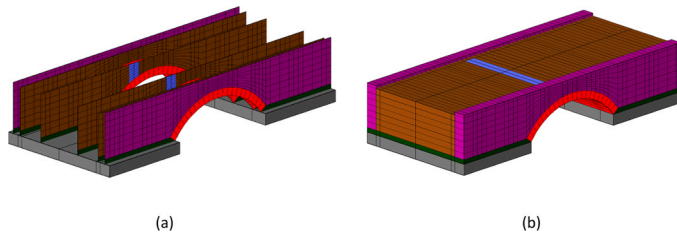


Fig. 7. Three-dimensional model with macro-elements along the transverse direction: (a) Numerical model; (b) Geometric model.

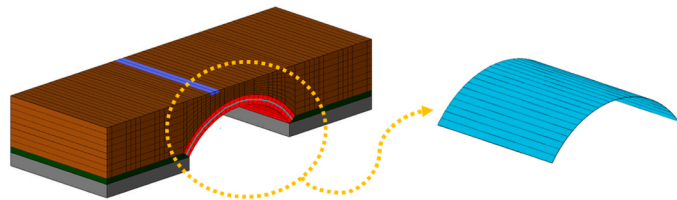


Fig. 8. View of the zero-thickness interface elements (strip model).

mechanism is modelled using an elasto-plastic constitutive law with fixed ductility. The yield domain is defined by a Mohr-Coulomb failure criterion, which captures the influence of both cohesion and internal friction. Similarly, the sliding failure mechanism along mortar joints or interfaces is governed by an elasto-plastic law, with a Mohr-Coulomb yield domain to account for shear resistance and slip due to frictional contact. These constitutive laws are applied not only to the masonry elements but also to the backfill material, with necessary adjustments made to account for the distinct mechanical properties and behavioural characteristics of soil.

A non-linear analysis is conducted on the bridges under the line loading using a classical non-linear incremental scheme. The analysis is performed using an iterative Newton-Raphson procedure with an energy convergence criterion, which considers the norm of the product of the residual unbalanced vector and the displacement increment vector. The magnitude of the increments is regulated by the arc-length algorithm, which effectively manages step size throughout the analysis.

The discretisation strategy includes standardising the number of rows of orthogonal interface springs to three in most models, as discussed in Section 2.1.1, with the macro-element size maintained below 30 cm to ensure sufficient detail in the numerical representation. Additional refinement of the mesh, including variations in the number of rows, is implemented in some models to achieve a more accurate description of localized behaviours, as discussed in subsequent sections. The maximum spring spacing adopted in the analyses is 10 cm.

The discretization of springs influences the accuracy of the results in a way analogous to the role of integration points in Gauss quadrature, i. e., more points can improve accuracy, but beyond a certain threshold, the improvement becomes marginal. Previous sensitivity analyses (e.g., [31]) have shown that for spring spacings between 5 and 15 cm, further refinement has negligible effect, supporting the chosen discretization.

4. Results

4.1. Single span bridges

This section focuses on the numerical validation of single-span MABs experimentally tested under various conditions, as reported in [38]. The analysis includes bridges with attached and detached spandrel walls and those with defects in arch rings, providing a comprehensive overview of different structural configurations. Specifically, the following models and their respective damages are considered: *Bridge 3.1* features detached spandrel walls, while *Bridge 3.2* includes both detached spandrel

walls and ring separation. *Bridge 3.3* represents a defect-free configuration, and *Bridge 3.4* exhibits ring separation. Additionally, *Bridge 5.1* has detached spandrel walls and *Bridge 5.2* has both detached spandrel walls and ring separation.

The numerical outcomes obtained in this study are also compared, where available, with the results from Zhang et al. [22], who adopted a mesoscale approach, and Gilbert et al. [45], who employed a Limit Analysis approach. These comparisons further validate the proposed methodology and highlight the consistency of the results across different modelling strategies.

4.1.1. Bridge 3.1 – detached spandrel walls

Bridge 3.1 is a 3 m-span MAB prototype with detached spandrel walls. The corresponding deflections at the intrados of the arch were measured by transducers placed at a quarter span. As discussed in the previous section, *Bridge 3.1* model employs an efficient strip model with only one set of elements along the width of the bridge specimen for the arch and the backfill meshes.

Two distinct models were developed for the analysis. The first model employs a coarse mesh with an element size of about 30 cm. This mesh configuration consists of 12 elements along the arch barrel and just one element along the thickness of the arch ring. The second model utilizes a more refined mesh, with an element size of about 10 cm, resulting in two macro-elements along the thickness of the arch ring and a total of 26 elements along the arch barrel, making up 52 elements. This refined mesh allows for a more detailed description of the multi-ring arch, which was constructed with stretcher bond.

The material properties specific to *Bridge 3.1* were not included in the study documented by [38]. Therefore, the material characteristics of *Bridge 3.3*, as detailed in Table 2, were considered for this model. This approach aligns with the methodology used in [22], where the material properties from *Bridge 3.3* were also utilized.

Fig. 9a illustrates the comparison between the experimental and numerical results in terms of global force-displacement plots, where the force is the total force applied and the displacement measured at the quarter span. In addition to the results from the present study, Fig. 9a also includes the numerical outputs reported by Zhang et al. [22] and Gilbert et al. [45].

A generally good agreement between the experimental and numerical response predicted by the strip-model of [22] is observed, with accurate reproduction of the initial stiffness, load capacity, and post-peak behaviour. Conversely, the results from [45], obtained using a Limit Analysis approach, underestimate the peak load by approximately 82%. A remarkable agreement can be observed between the experimental and DMEM numerical outcomes, with no significant discrepancies between the results obtained with the coarse and finer mesh. This is a very interesting result, showing that the proposed DMEM strategy exhibits reduced sensitivity to the mesh size, unlike other modelling approaches that suffer from localisation issues when a softening material behaviour is employed. A very good agreement is also observed in terms of collapse mechanisms (Fig. 9b), thereby confirming the robustness of the findings. It is also worth to note that the DMEM analysis converged in only 15 s for the coarse mesh and 50 s for the finer mesh on a standard desktop.

Fig. 10a-b illustrates the contour plots of the plastic work, obtained by [22], and of the normal stresses in the arch barrel, obtained using DMEM model with refined mesh. Both figures indicate the presence of four radial cracks contributing to the collapse mechanism. The normal stresses are linearly distributed along the arch barrel thickness. The absence of any discontinuity in stress at the interface between the arch rings suggests that there is no sliding between the two rings of the arch.

Fig. 11 illustrates the stress contours for the principal compressive stress component. Initially, for small loads, a stress bulb following Boussinesq's elastic stress distribution pattern is observed beneath the load. As the load intensifies, the soil under the load begins to yield and to exhibit plastic behaviour. As the shear stresses reach their limiting value

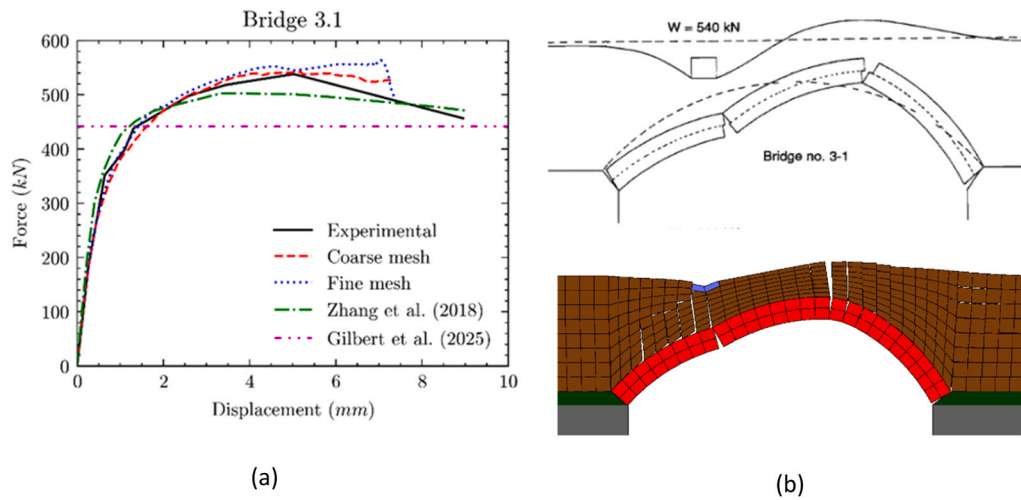


Fig. 9. Experimental-numerical (a) load-displacement response and (b) collapse mechanism (deformed displacement scaled by 10) for Bridge 3.1 [38].

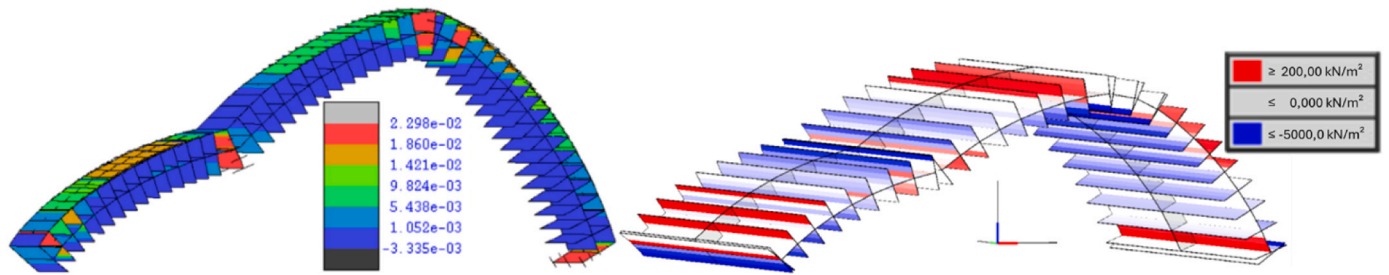


Fig. 10. Deformed shape of the arch barrel with highlighted (a) Plastic work [22] and (b) Normal stress at interfaces between individual masonry units for Bridge 3.1.

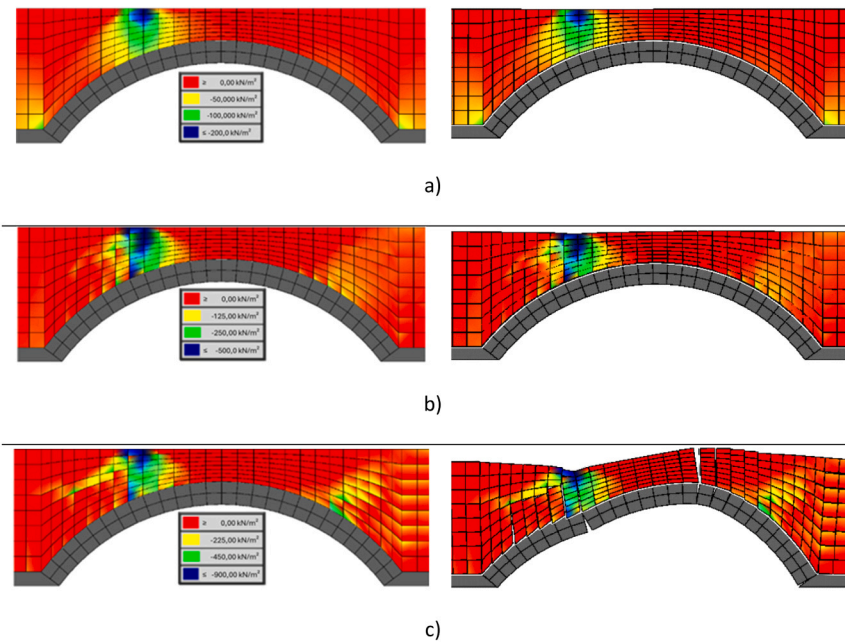


Fig. 11. Principal compression stress in backfill: left column (undeformed), right column (deformed, displacement scaled by 10). Load cases: a) 120 kN, b) 400 kN, and c) 560 kN.

at the interfaces between the elements representing the soil, the normal stresses in the vertical direction under the loading tend to localise in a narrower band. Despite these changes, the soil continues to effectively

distribute the load. As the arch undergoes significant deflections, the upward displacement of the arch barrel at three quarter of the span induces notable stresses in the soil, with the soil fill providing passive

resistance. The stresses related to the passive soil resistance are particularly pronounced in Fig. 11-c when the load approaches its maximum value.

In order to obtain the colour map reported in Fig. 11, the principal compressive stress field must be reconstructed by applying the following procedure:

1. A grid of points is defined for each element (using either equally spaced or Gauss schemes, as commonly adopted in Finite Element software).
2. For each grid point, its projections onto the interfaces connecting the element with adjacent ones are determined.
3. The links closest to these projected points on the interfaces are selected to reconstruct the stress tensor.
4. The normal and tangential stresses associated with each selected link are weighted according to the distance of the grid point from the corresponding interface.
5. The approximate stress tensor at the grid point is obtained by summing the contributions of all relevant links.
6. Finally, a colour map is generated by interpolating the values across the grid of points.

4.1.2. Bridge 5.1 – detached spandrel walls

This section presents the results obtained for Bridge 5.1, a 5 m-span MAB prototype with detached spandrel walls. Fig. 12a illustrates the global force-displacement plots for the experimental and DMEM numerical models, revealing a significant discrepancy in terms of stiffness, but a good agreement in terms of peak strength. In addition to the results of the present study, Fig. 12 also includes the numerical output from [45]. In this case, the Limit Analysis approach overestimated the peak load of approximately 10%. According to [45], this over-prediction can be attributed to the sudden onset of partial ring separation during the experimental load test, which reduced the actual capacity of the bridge specimen.

It must also be observed that the experimental findings for Bridge 5.1 [38] highlighted an unexpected high level of friction between the backfill and the detached spandrel/wing walls. This unforeseen friction likely contributed to increase the structural stiffness. It was noted that the interaction between the soil and structural elements was so pronounced that it resulted in the elevation of a portion of the spandrel walls, distant from the load application point, by several millimetres [38]. This movement led to visible cracking at the base of these walls when subjected to a load of 1620 kN. Thus, the spandrel walls interacting with the backfill have likely played a significant role in the behaviour of the structure, particularly during the early stages.

In order to simulate this interaction between the backfill and spandrel walls, a revised model was developed, incorporating explicitly the detached spandrel walls, connected only to the backfill, as shown in Fig. 12b. This updated model demonstrates improved consistency between experimental data and numerical predictions, as demonstrated in Fig. 12a. Nonetheless, the stiffness observed in the tests remains higher than the simulated one. The revised model simulation required about 720 s of CPU time, while the strip-model simulation required about 360 s.

Even though the revised model for Bridge 5.1 incorporates the spandrel walls, it is crucial to recognize the limitations of the modelling strategy when interpreting the discrepancy in stiffness values between the numerical results and experimental outcomes. This difference may be justified by observing that the current version of the HiStrA software does not account for the confinement effect of Poisson's ratio provided by the spandrel walls to the fill material. This confinement can significantly increase the stiffness of the fill material and enhance the friction between the soil and spandrel walls. Consequently, the absence of this effect in the current macro-element approach likely contributed to the observed differences in stiffness, highlighting an area for potential refinement in future modelling efforts to achieve more accurate alignment with experimental findings. This particular issue was also highlighted in the study documented in [35], which compared the results from a 3D non-linear FEM simulation against those obtained from a DMEM simulation.

4.1.3. Bridge 3.2 – detached spandrel walls and ring separation

This section presents the results obtained for Bridge 3.2, a 3 m-span MAB prototype with detached spandrel walls and ring separation ('damp sand' interface). The study reported in [38] did not provide the masonry properties for Bridge 3.2. Therefore, the material specifications considered for Bridge 3.3, as outlined in Table 2, were adopted in the numerical analysis. The results presented in Fig. 13a illustrate a very good agreement between the experimental and numerical results. The collapse mechanisms, illustrated in Fig. 13b, further confirm this perfect agreement, thus demonstrating the effectiveness of the DMEM in accurately capturing the effect of defects in the arch rings. Including the friction interface increases the run-time to 80 s, compared to the 50 s taken for Bridge 3.1.

Fig. 14a illustrates the plastic work, obtained by [22] and shown here for comparison, within the arch barrel, while Fig. 14b details the normal stress profile. Both figures display the deformed shape of the arch and the development of four radial cracks that play a significant role in the collapse mechanism, exhibiting similar behaviours. In contrast to the observations in Fig. 10, a non-linear stress distribution is

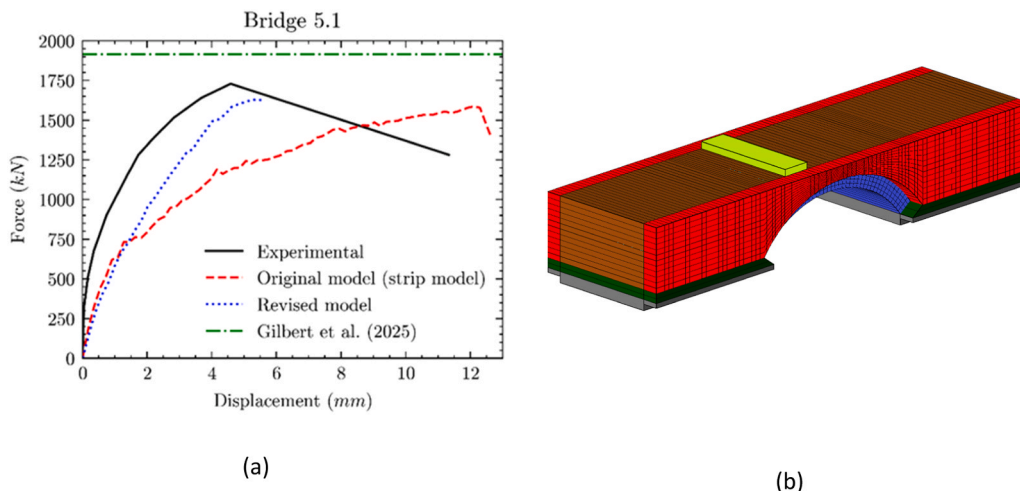


Fig. 12. (a) Experimental-numerical load-displacement response and (b) revised numerical representation of Bridge 5-1.

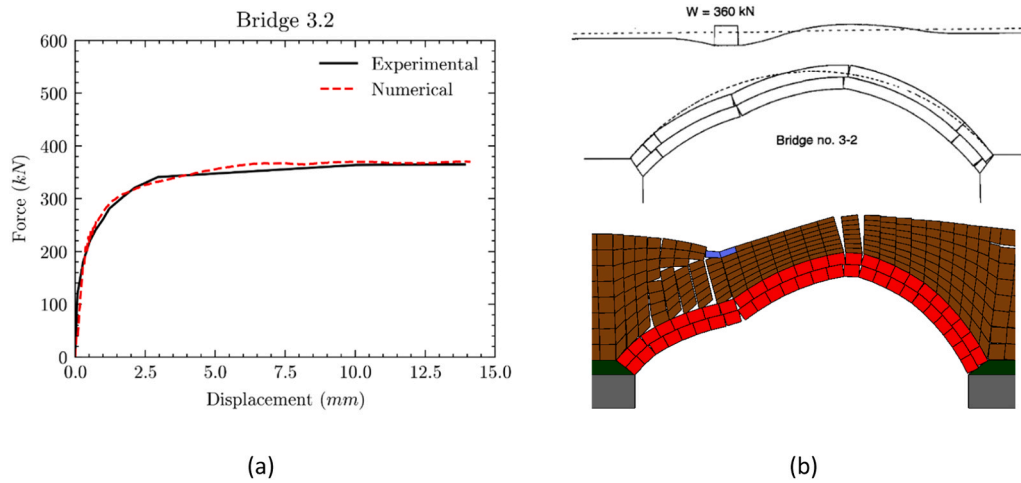


Fig. 13. Experimental-numerical (a) load-displacement response and (b) collapse mechanism (deformed displacement scaled by 10) for Bridge 3.2 [38].

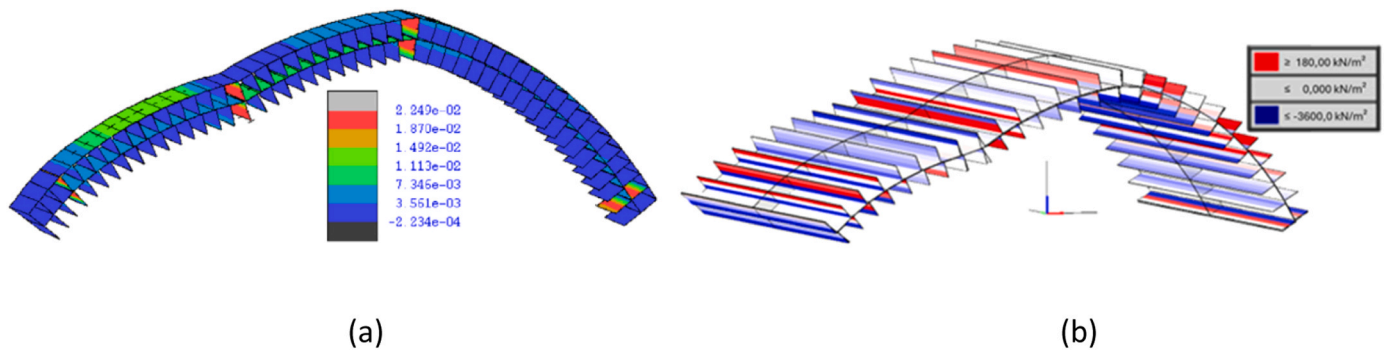


Fig. 14. Deformed shape of the arch barrel with highlighted (a) Plastic work [22] and (b) normal stress at interface between individual masonry unit for Bridge 3.2.

evident between the arch rings in Fig. 14, underscoring the pronounced discontinuities at the interfaces due to ring separation. Each ring displays independent behaviour, characterized by localized compression and tension within individual rings. Conversely, the stress distribution in Fig. 10 shows a more collaborative interaction among the rings, indicating effective load sharing and cooperative structural behaviour across the entire arch.

A comparative analysis of the load-displacement responses from

Fig. 9a and Fig. 13a, representing models with damaged and undamaged circumferential mortar joints, respectively, reveals significant differences. The load capacity in the model with damaged joints (Bridge 3.2) is markedly lower, approximately 360 kN, compared to about 540 kN in the reference model (Bridge 3.1). This substantial reduction in load capacity underscores the critical impact of masonry defects on the structural performance and stability of the bridge under vertical loads.

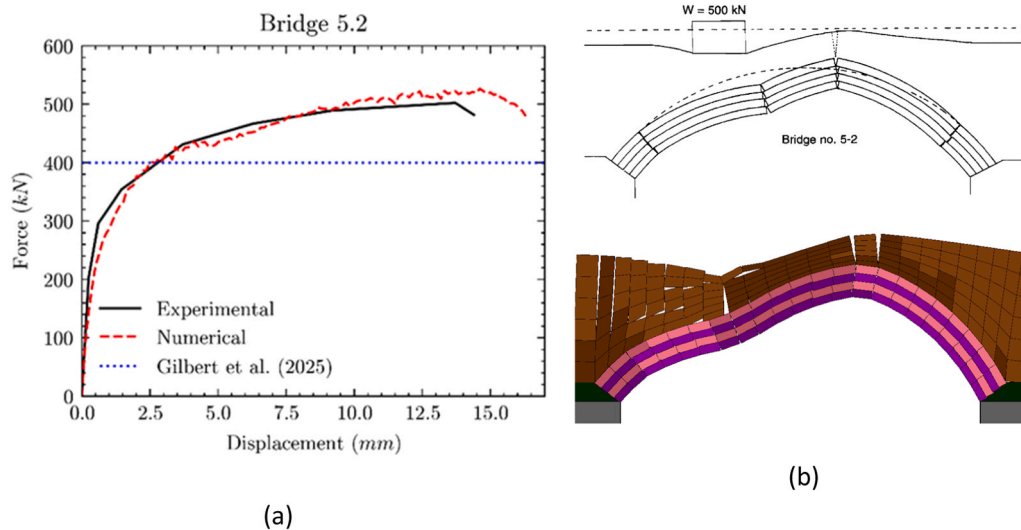


Fig. 15. Experimental-numerical (a) load-displacement response and (b) collapse mechanism (deformed displacement scaled by 20) for Bridge 5-2 [38].

4.1.4. Bridge 5.2 – detached spandrel walls and ring separation

Bridge 5.2 is a variant of Bridge 5.1, specifically designed to include defects at the interfaces between the arch rings. Fig. 15 compares Bridge 5.2's response according to the experimental test and the numerical simulation, showing one more time that the proposed DMEM approach provides very accurate response estimates. Computational cost remains modest, with a total run-time of 360 s. Fig. 15a also includes the numerical output from [45]. In this case, the Limit Analysis approach underestimates the peak load at about 80% of the expected value. This underestimation may highlight the limitations of the method in capturing the complex interaction between defected arch rings.

When comparing the load-displacement responses illustrated in Fig. 12a and Fig. 15a, which represent models with intact and damaged mortar joints, respectively, there is a notable discrepancy. The load capacity of Bridge 5.2, with its compromised joints, is significantly reduced to about 500 kN, in contrast to approximately 1720 kN in the undamaged Bridge 5.1, implying a reduction of 71%. This reduction is significantly larger than the 33% reduction observed between Bridges 3.1 and 3.2. This can be explained by noting that Bridge 3.2 had only one defective interface compared to the three in Bridge 5.2.

The comparative analysis of Bridges 5.1 and 5.2 in Fig. 16 reveals significant differences also in the stress distribution within the arch barrels. In Bridge 5.2, where the arch rings are defective, there is a noticeable reduction of shear transfer, with each ring behaving almost independently and showing localized compression and tension. Conversely, the stress profile in Bridge 5.1 exhibits a cooperative interaction among the rings, indicating effective load sharing and monolithic structural behaviour across the arch. This underscores the critical role of interface integrity in the overall stability and performance of masonry arch bridges. Additionally, Bridge 5.2 displays characteristics similar to those observed in Bridge 3.2, such as lower peak loads and discontinuities between the rings.

4.1.5. Bridge 3.3

Bridge 3.3 is a model featuring attached spandrel walls. Due to the lack of specific material properties for the spandrel walls in [38], parameters of the Drucker-Prager model used for the spandrel walls in Zhang et al. [22] were adopted. These included a relatively low Young's modulus of 4 GPa, a cohesion value of 0.20 MPa, and a friction angle of 30°.

The Mohr-Coulomb yielding criterion was here employed by assuming the same values of cohesion and friction angle. Since no experimental evidence of relative sliding between the two rings of the arch barrel was observed, in order to speed up the numerical analysis, in this case a single layer was employed to model the arch.

Fig. 17a displays the force-displacement plots according to the experimental test and the numerical analyses carried out in this study and in Zhang et al. [22]. While there is a minor discrepancy in stiffness values, there is a strong alignment in peak strength between our results and those of Zhang et al. [22]. However, Zhang et al. [22] did not

provide data beyond the peak load, preventing any comparison of post-peak behaviour. It should also be highlighted that the bridge model featuring attached spandrel walls demonstrated considerably greater stiffness and peak load capacity than the model with detached spandrel walls (Bridge 3.1). Notably, the peak load in the attached spandrel wall model exceeded that of the detached model by 25%. The 3-D model takes 560 s (i.e., approximately 9 min) to run.

The results shown in Fig. 17 refer to the model described in Fig. 7. A mesh sensitivity analysis was performed to assess the adequacy of the adopted transverse discretisation. In particular, the backfill and spandrel walls were discretised along the width direction using an increased number of macro-elements, while keeping all material properties unchanged. The comparison of the corresponding load-displacement responses and deformation patterns showed only marginal differences with respect to the reference model. No significant variation in peak load, initial stiffness, or collapse mechanism was observed. These results indicate that the adopted transverse discretisation is sufficient to capture the global structural response of the bridge with attached spandrel walls, while maintaining a reduced computational cost.

It is observed that the results of the numerical models are not fully aligned with the experimental outcomes, the latter exhibiting a lower peak capacity. It is noteworthy that [38] noted that bridges featuring attached spandrel walls (specifically Bridges 3.3 and 3.4) exhibited cracking, at notably low loads, between the extrados of the arch barrels and the attached spandrel walls. This cracking occurred before any cracking within the arch barrel, associated with hinge formation (at 280 kN and 200 kN for Bridges 3.3 and 3.4, respectively). This may explain the discrepancy between the numerical and experimental results.

Significant plastic deformations, indicative of crack formation, are observed in the spandrel walls of the bridge model (Fig. 18). The highest deformations are found in locations that match those where cracking was observed in the experimental tests.

Fig. 19 illustrates the displacements at the quarter span across the arch width under various load levels. With an increase in load, the distribution of displacements becomes more uneven. Displacements near the spandrel walls are significantly reduced compared to those at the centre of the arch. This pattern results from transverse effects and interactions with the side walls, which cause the central section of the arch barrel, furthest from the walls, to experience greater displacement. The displacement profile at peak load is very similar to the one obtained by Zhang et al. [22].

4.1.6. Bridge 3.4 – ring separation

Bridge 3.4 is characterised by attached spandrel walls, as is Bridge 3.3, but has a weak connection between rings. Therefore, the model of Bridge 3.4 is consistent with that of Bridge 3.3, with the notable modification of incorporating the 'damp sand' interface between the arch rings. It is noteworthy that, contrary to expectations, the maximum load capacity observed during the experimental campaign for Bridge 3.4 (with

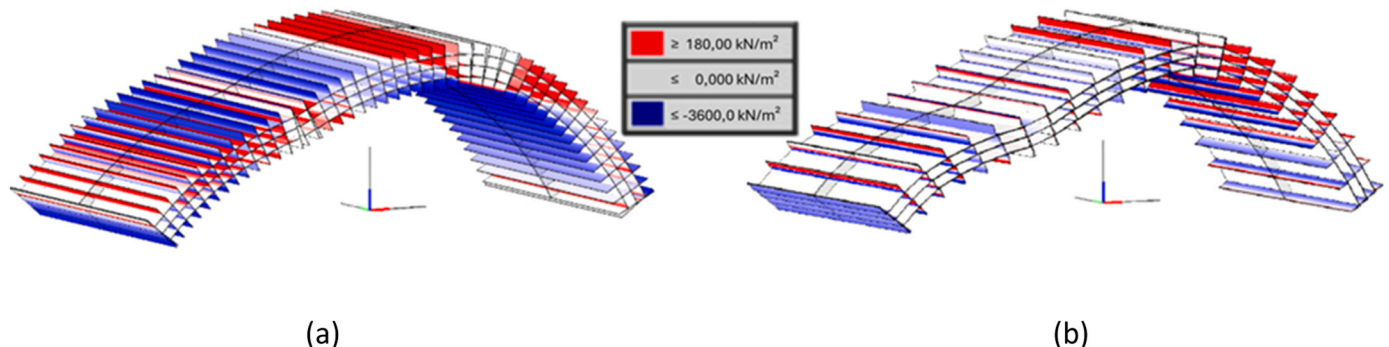


Fig. 16. Deformed shape of the arch barrel with highlighted normal stress at interface between individual masonry unit for (a) Bridge 5.1 and (b) Bridge 5.2.

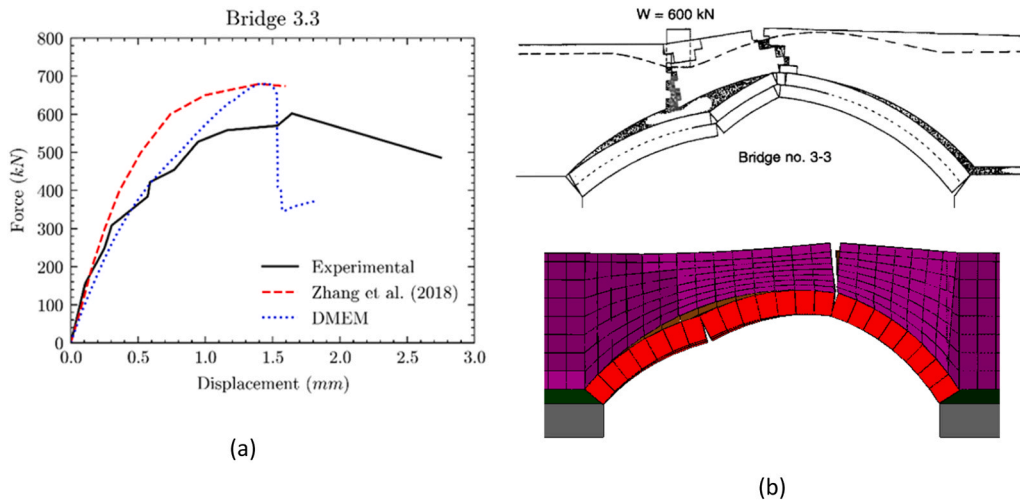


Fig. 17. Experimental-numerical (a) load-displacement response and (b) collapse mechanism (deformed displacement scaled by 40) for Bridge 3.3 [38].

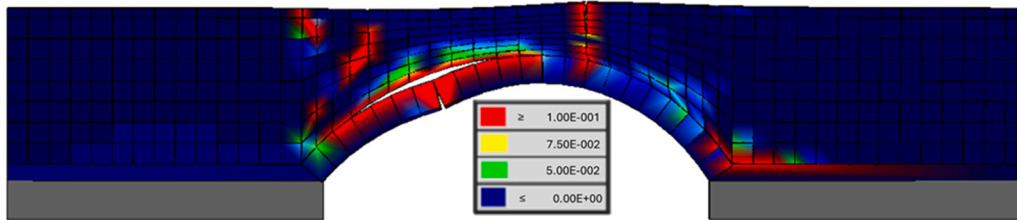


Fig. 18. Deformed shape with highlighted ratio plastic strain for Bridge 3.3.

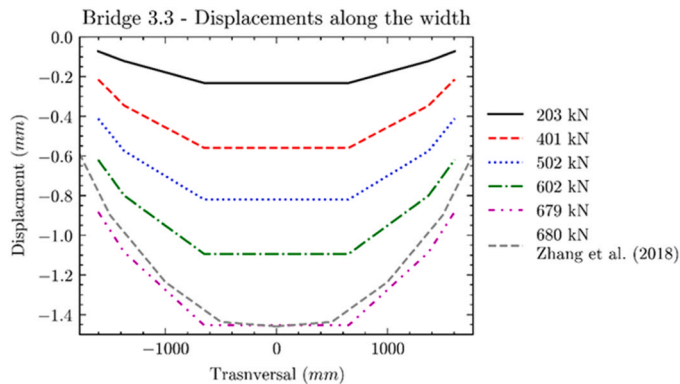


Fig. 19. Displacements along the arch width at different loading levels for Bridge 3.3.

attached spandrel walls and ring separation) was lower than that recorded for *Bridge 3.2* (with detached spandrel walls and ring separation), see [Table 1](#). This reduction in structural integrity can be attributed to the premature cracking observed in *Bridge 3.4* under relatively low loads, as discussed in [38].

[Fig. 20](#) illustrates the global force-displacement plots for the experimental and numerical model of *Bridge 3.4*, revealing a significant discrepancy in terms of peak load, but a good agreement in terms of initial stiffness. Because of the finer mesh and weak-interface links, the solution took around 1800s (i.e., approximately 30 min) to complete.

For comparative analysis, the results from *Bridge 3.2* (scaled by the ratio between the width of *Bridge 3.2* and that of *Bridge 3.4* to simulate the same effective width as *Bridge 3.4*) and *Bridge 3.3* (attached spandrel walls and no ring separation) are plotted in the same figure. It can be observed that the curve of *Bridge 3.4* lies between the curve of adjusted

Bridge 3.2 and *Bridge 3.3*. The load capacity of *Bridge 3.4*, compromised by defective ring interfaces, is reduced to about 522 kN compared to approximately 680 kN for the undamaged *Bridge 3.3*, representing a 23% reduction. This reduction is broadly consistent with the reduction observed in the capacity of *Bridge 3.1* compared to *Bridge 3.2* (around 33%), demonstrating a consistent impact of structural defects across different bridge configurations.

[Fig. 21](#) presents the transverse displacement profile across the arch width at the quarter span for *Bridge 3.4*. As the load increases, the central portion of the arch barrel undergoes greater vertical displacement, whereas areas near the spandrel walls exhibit significantly lower movement. This uneven distribution is attributed to the stiffness provided by the spandrel walls. The same behaviour is observed in [Fig. 19](#), confirming that the presence of spandrel walls significantly influences the structural response by promoting a spatially non-uniform displacement profile. These response highlights the necessity of incorporating transverse discretisation and three-dimensional modelling to accurately capture the interaction between the arch barrel and spandrel walls and to ensure realistic simulation of the bridge's structural behaviour.

4.2. Multi-span bridges

This section presents the numerical validation of the experimental tests on multi-span masonry arch bridges described in [39]. The analysis encompasses bridges featuring attached and detached spandrel walls to examine their influence on the structural behaviour of the bridges. The experimental setup included testing three large-scale model bridges with three spans measuring 3 m each. The geometrical design of these spans mirrored that of the 3 m single-span bridges previously examined [38]. *Bridge MS1* and *Bridge MS3* were constructed with spandrel walls directly connected to the arches but subjected to different loading conditions. Conversely, *Bridge MS2* featured spandrel walls not attached to the arches across its spans. Again, since no defect was introduced at

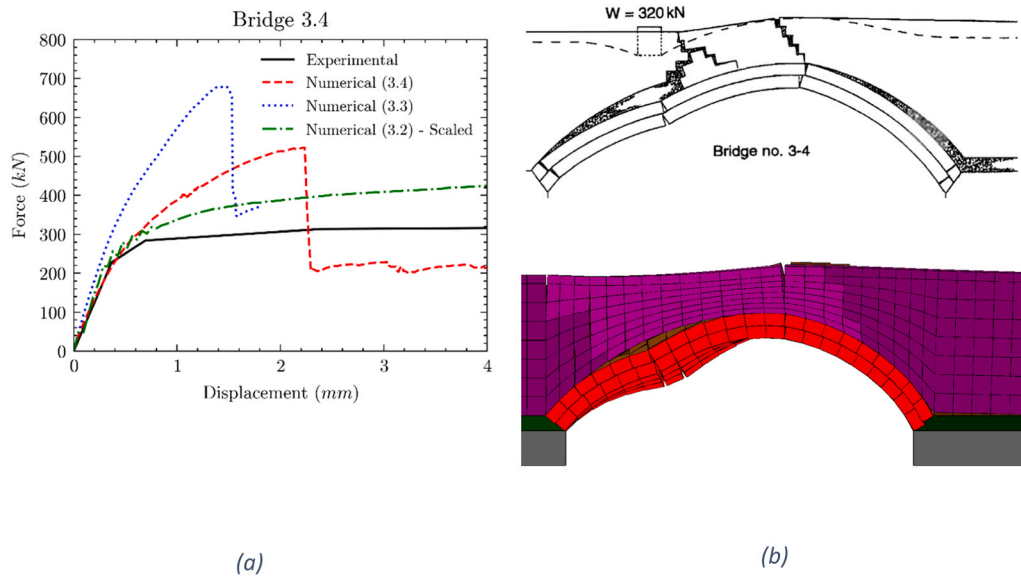


Fig. 20. Experimental-numerical (a) load-displacement response and (b) collapse mechanism (deformed displacement scaled by 90) for Bridge 3.4 [38].

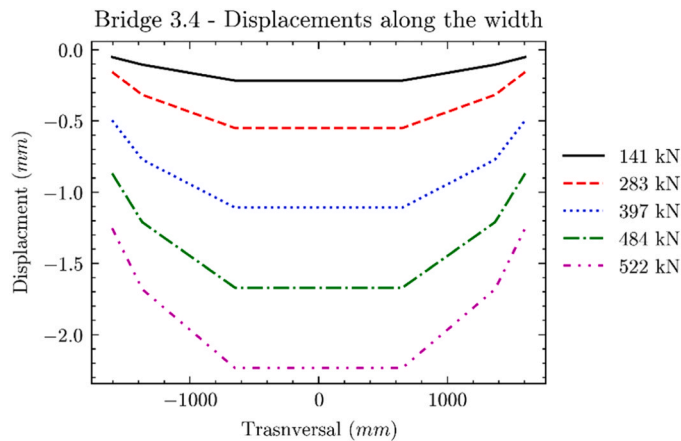


Fig. 21. Displacements along the arch width at different loading levels for Bridge 3.4.

the interface between the two rings of the arch, a single layer was employed to model the arch in the numerical simulations. Each multi-span analysis was completed in 300 s, even with the three-stage load sequence (*Bridge MS3*), as better discussed in Section 4.2.2.

In addition to the numerical results obtained with the proposed DMEM approach, the outcomes were also compared with other numerical predictions available in the literature, namely: Tubaldi et al. [23], who employed a FEM mesoscale modelling strategy; Zizi et al. [21], who adopted a FEM approach with masonry and backfill interaction modelled as friction contact interaction; and Gilbert et al. [45], who applied a limit analysis approach with rigid block modelling strategy. These comparisons are included in the subsequent sections, where applicable, to provide a broader context for the validation of the DMEM approach against other state-of-the-art numerical modelling techniques.

4.2.1. Bridge MS2 – detached Spandrel Walls

Bridge MS2 is modelled by employing a simplified strip model approach consistent with previous analyses of similar configurations (Section 4.1). Fig. 22 presents the comparative responses for the experimental and numerical results. In Fig. 22a, the load-displacement curve for the arch demonstrates a strong correlation between the two, capturing the key trends and non-linearities in the response. Fig. 22a

also includes the numerical outputs from other authors. Tubaldi et al. [23] provided a simulated response that is quite close to the experimental one, with good agreement in terms of initial stiffness, load capacity, and post-peak response. Similarly, Zizi et al. [21] produced a numerical curve whose consistency with the experimental response can be appreciated. Gilbert et al. [45], using a Limit Analysis approach, achieved an excellent match with the experimental peak load, accurately reproducing the structural response of the bridge.

Similarly, Fig. 22b illustrates the vertical load-horizontal displacement responses of the piers at the north and south ends, showing excellent agreement between experimental and numerical data. Fig. 22b also includes the numerical results from Zizi et al. [21], which demonstrate good agreement with the experimental response. Furthermore, the analysis of collapse mechanisms (shown in Fig. 22c) for both the experimental and numerical investigations display a significant degree of similarity, confirming the validity and reliability of the DMEM strategy for the analysis of bridges with single and multiple spans.

Fig. 23a illustrates the contour plot of the damage in tension as reported in Tubaldi et al. [23] and Fig. 23b illustrates the plastic strains under tension, providing insight into the progression of cracking within the arch barrel. It is noted that damage extends to the arch of the first span despite the absence of major cracks in Fig. 22. These zones of high plastic deformations agree with the location where cracks have been identified.

4.2.2. Bridge MS1 and bridge MS3 – attached Spandrel Walls

Bridge MS1, featuring attached spandrel walls, has the same configuration as *Bridge MS2*, with the addition of the spandrel walls. Fig. 24 reveals substantial differences between numerical predictions and experimental observations. Notably, the experimental study [39] highlighted significant discrepancies in the readings obtained from two load cells, suggesting potential inaccuracies in load measurement due to the cells' sensitivity to eccentric loading conditions. Consequently, the load value associated with the bridge's failure is approximate. Although a direct comparison of the bridge's load capacity is not feasible, it is observed that the initial load-displacement response in the experimental data aligns closely with the numerical model. This alignment indicates that the model successfully captures the bridge's initial stiffness, providing a reliable representation of its early-stage behaviour.

Bridge MS3 exhibited a different load application arrangement. Rather than at the quarter span, the load was applied nearer the centre, 240 mm north of the midpoint. This initial loading strategy resulted in

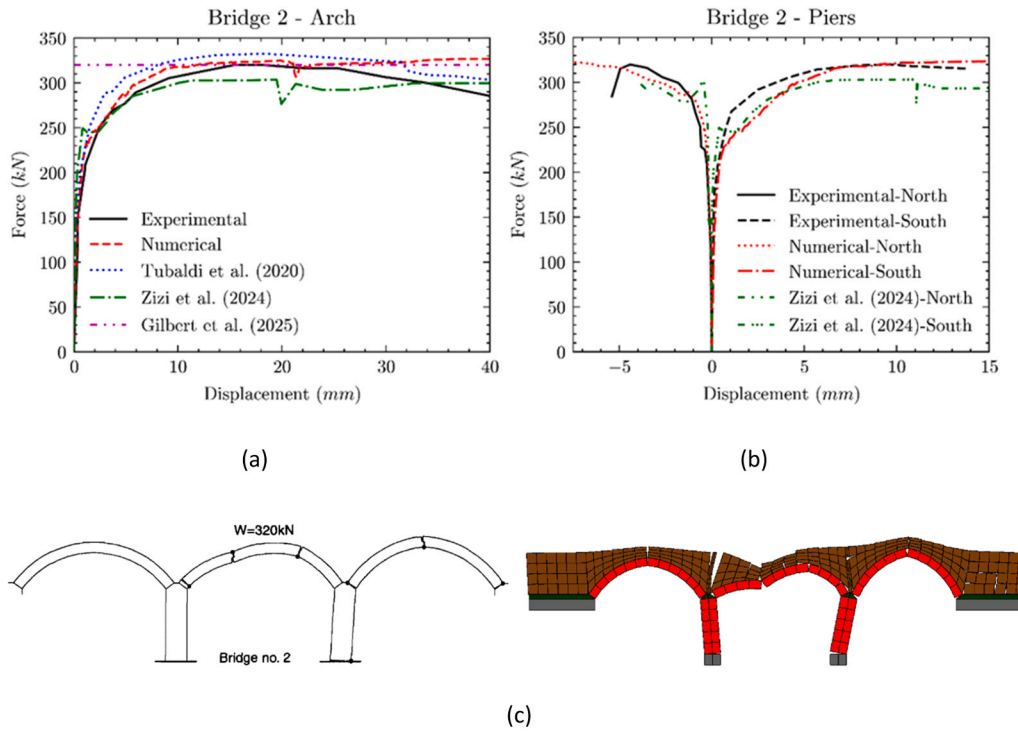


Fig. 22. Experimental-numerical (a) arch and (b) pier load-displacement response and (c) collapse mechanism (deformed displacement scaled by 5) for Bridge MS2 [39].

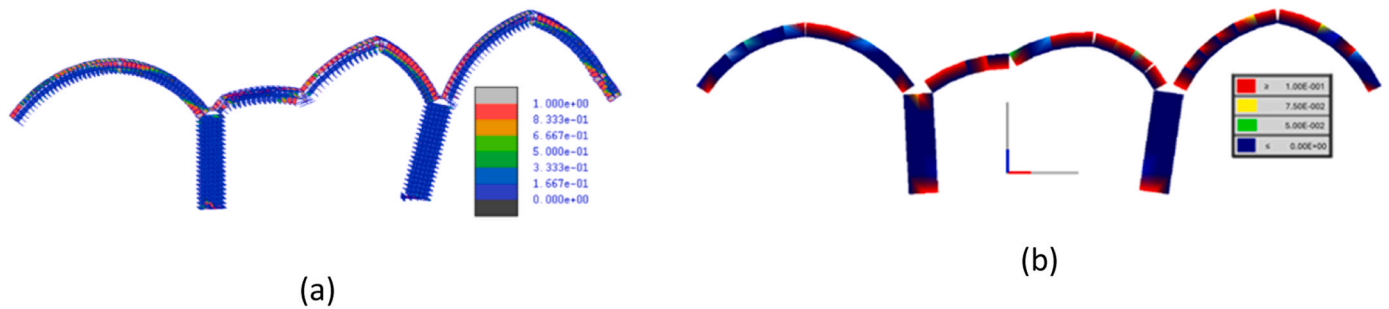


Fig. 23. Deformed shape of the arch barrel with highlighted (a) damage parameter [23] and (b) tension plastic strain for Bridge MS2.

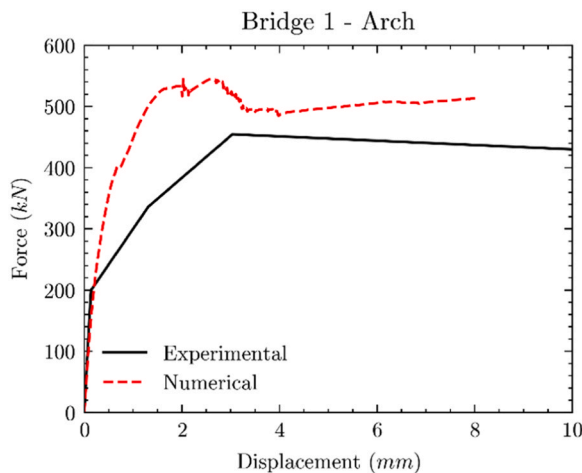


Fig. 24. Experimental-numerical pier load-displacement responses for Bridge MS1.

greater displacements of the northern pier, which was not desired. To achieve a more pronounced movement in the southern span and encourage failure through a mechanism involving this area, four concrete blocks with a combined weight of 48 kN were placed at the top of the northern span. Due to this new load configuration, the numerical assessment has been carried out in three distinct stages. The first stage has involved applying the bridge's vertical weight. The second stage has involved the addition of the extra weight at the midpoint of the northern span. Finally, the third stage has simulated the application of the downward force. This methodology mirrors that of a previous study [21], which adopted a value of the friction angle of the backfill equal to 30° . This value was also adopted in the analyses to obtain comparable results.

The results, illustrated in Fig. 25, display the force-displacement curve and damage patterns. While there is a very good agreement in the initial stiffness between the experimental and numerical results, significant differences emerge around the 200 kN load. The two numerical models (DMEM and FEM [21]) show close agreement up to the peak load; however, divergences appear after that. Notably, the FEM model exhibits an abrupt decrease in load capacity post-peak. In contrast, the DMEM model demonstrates a more gradual plastic

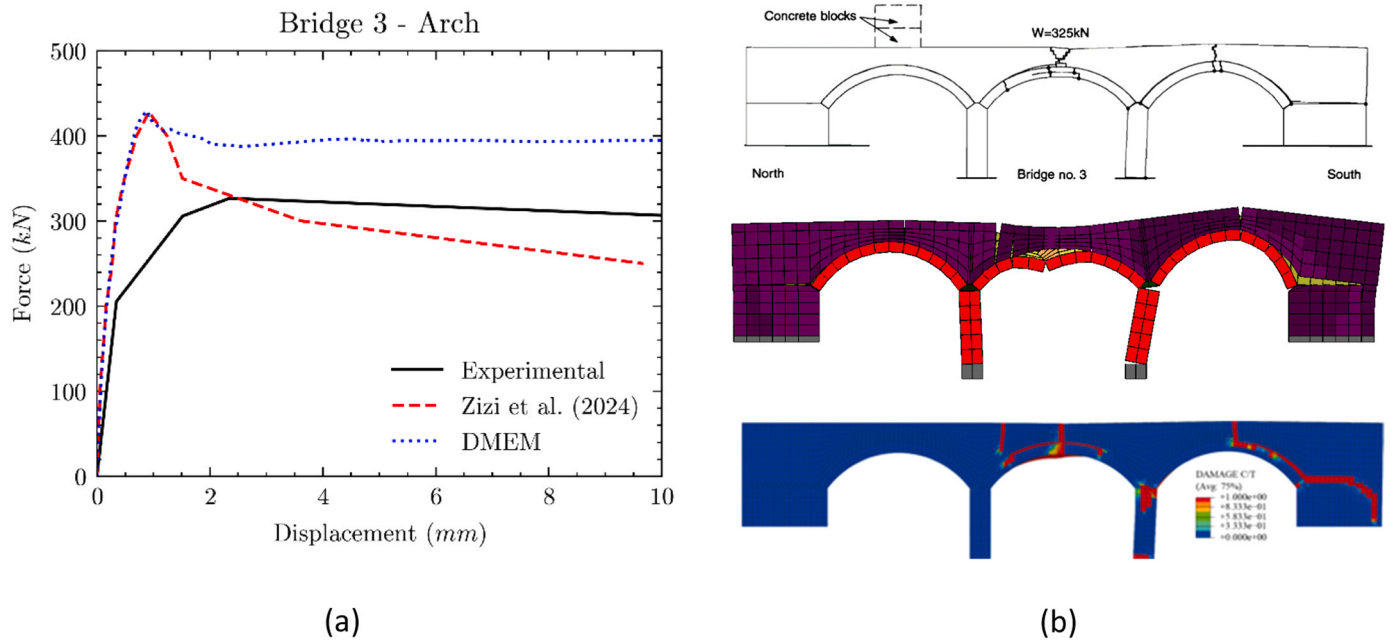


Fig. 25. Experimental-numerical (a) load-displacement response and (b) collapse mechanism (deformed displacement scaled by 50) for Bridge MS3 [21].

deformation. This discrepancy may be attributed to the differences in post-peak material properties, which were not specified in the experimental study. Furthermore, the agreement between the experimentally observed and numerically predicted failure modes and cracking patterns (Fig. 25b) provides validation for the DMEM modelling strategy.

5. Discussion

The results demonstrate the reliability of the proposed model and relevance of the features updated in this study, that allowed extending the DMEM strategy to MABs. Specifically, this upgraded macro-element configuration can accommodate interfaces not only along the perimeter but also on the top and bottom faces (i.e., on all six faces), which is necessary to capture not only the spatial structural behaviour of massive structures, but also load diffusion in the backfill and the influence of spandrel walls.

One critical aspect of the modelling strategy for MABs is the backfill. Its primary roles are the diffusion of vertical loads and providing passive thrust where the arch uplifts. In the DMEM, passive and active soil pressure effects arise from the explicit modelling of the backfill. The passive thrust effect is illustrated in Fig. 11c, where the opening of a hinge at the extrados of the right half of the arch is delayed due to the backfill's resistance, which is reflected in the pressure shown in the colour maps. Regarding load diffusion, the typical stress field of loaded soil is shown in Fig. 11a. While the backfill model could be enhanced using a more sophisticated constitutive behaviour, the adopted simple laws are consistent with the DMEM philosophy, that is a simple approach is acceptable as long as it provides satisfactory results while keeping low the computational demand. Indeed, more complex laws can be implemented as needed.

To properly contextualize the proposed DMEM approach, it is important to highlight its differences, advantages, and limitations compared to other numerical strategies. With regard to the computational burden, the numerical campaign underscores the remarkable computational efficiency of the DMEM. Even the most demanding three-dimensional models, characterised by fine discretisation and frictional ring-separation interfaces, converged in only a few minutes on a modern laptop equipped with a 13th-Gen Intel® Core™ i7-13650HX processor and 32 GB of RAM, despite the fully nonlinear formulation. Consequently, DMEM achieves predictive accuracy at a fraction of the CPU

time required by high-fidelity FEM or DEM analyses, which often run for several hours or even days. This efficiency makes exhaustive parametric studies and routine structural assessments of masonry arch bridges entirely feasible on readily available hardware.

However, a direct comparison between DMEM and other fast tools available in the literature, such as those based on limit analysis, requires a careful evaluation of the respective approaches. For example, a comparison with the limit analysis method cannot be made solely in terms of computational cost, as their underlying features differ significantly. Limit analysis is a very fast tool that primarily provides information on the load multiplier and the collapse mechanism of a masonry structure. For multi-span masonry arches (MABs) or arched structures in general, applications of limit analysis are typically planar [46–48], including approaches that explicitly address support/pier settlements through alternative kinematic descriptions [49], while only recently three-dimensional applications have been developed [18,50,51]. Limit analysis typically employs simple constitutive laws that cannot capture the softening behaviour of masonry under compression or tension. Passive and active soil pressures are represented through calibrated restraint elements [45], while backfill and spandrels are approximated as loads rather than explicitly modelled. Planar models are justified via the concept of effective width [45]. On the contrary, DMEM adopts a structural analysis approach, supporting nonlinear static and dynamic simulations that provide detailed information on displacement capacity and damage progression. Consequently, direct comparisons of computational performance between DMEM and limit analysis are inappropriate, as the two methods produce fundamentally different outputs.

A key advantage of DMEM is its ability to capture the full three-dimensional geometry and behaviour of MABs. Including spandrels enables accurate 3D response, and the model also accommodates eccentric vertical loads and curved bridge geometries. Explicit modelling of backfill removes the need for load diffusion approximations or effective width concepts, ensuring reliable 3D structural behaviour.

For all the mentioned reasons, the primary aim of this study is not to compare DMEM with limit analysis, but rather with DEM and FEA, as the proposed method provides information on behaviour far from collapse. In this respect, the model yields accurate response estimates, with error levels comparable to mesoscale FEA results. Where results from alternative strategies were available, comparisons show similar levels of accuracy. Notably, some of the simulated tests had not been considered

in other studies using alternative modelling approaches.

The application of DMEM to nonlinear modelling of MABs is still at an early stage, particularly regarding soil–structure interaction. Nevertheless, DMEM offers distinct advantages over both FEA and DEM. Compared to FEA, it significantly reduces computational demand by limiting degrees of freedom and employing uniaxial constitutive laws. At the same time, its use of deformable elements connected by discrete interfaces allows, in certain cases, to outperform DEM and to operate at larger element scales. Despite the need for further studies on DMEM's capability to model soil–structure interaction, the present study represents an essential step toward analysing specific phenomena affecting MABs.

6. Conclusion

This study presents the results of an extensive numerical campaign evaluating the capabilities of the Discrete Macro-Element Modelling (DMEM) method in assessing the structural behaviour of Masonry Arch Bridges (MABs) under vertical loads applied at deck level. Comprehensive validation against experimental and numerical benchmarks has proven DMEM as a powerful and computationally efficient tool for evaluating MAB performance across diverse geometries, damage states, and loading conditions. The presented study represents a significant advancement both from a computational perspective and in terms of novel applications. Specifically, the ability to accommodate interfaces on all six faces of the element enables, for the first time, the modelling of massive structures where all three spatial dimensions are comparable. As a result, the nonlinear modelling of masonry arch bridges (MABs) becomes feasible. An extensive validation campaign has been carried out and is presented as part of this study.

Notably, DMEM has provided accurate estimates of initial stiffness for all bridge models studied, except for Bridge 5.1, likely due to specific features of the experimental setup. Furthermore, DMEM has accurately predicted the load capacity of MABs, with errors comparable to those obtained using more computationally intensive modelling methods. Although discrepancies were observed in simulating the post-peak behaviour of the tested prototypes, it is important to recognize that available experimental data from material and component tests lacked sufficient information to define masonry's post-peak behaviour under tension or compression.

The findings illuminate the method's strengths in simulating nonlinear behaviours and critical failure mechanisms with remarkable precision. DMEM's adaptability to various bridge typologies underscores its utility as an indispensable resource for engineers tasked with the conservation and maintenance of architectural heritage.

It is worth noting that the examples considered in this study primarily focus on square bridges and segmental circular arches. However, other configurations, such as skew spans and non-circular arches, remain unexplored in this context and warrant further investigation. The behaviour of such structures may introduce additional complexities that could affect both the accuracy of DMEM predictions and the computational cost of the method.

One important extension of the current version of the macro-element will be the inclusion of the Poisson effect, which is likely to have some impact on the interaction between the spandrel walls and the backfill. This extension will enable the investigation of masonry arch bridge configurations in which confinement and dilatancy effects govern the backfill behaviour and the backfill–spandrel interaction mechanisms. The potential impact of such advancement is significant, as it could lead to the development of a more robust and universally applicable tool for the structural assessment of MABs.

Additionally, it is essential to extend the application of the DMEM beyond the current case studies to include a broader range of loading scenarios. This includes specifically investigating the behaviour under moving traffic loading, the impact of foundation settlements on bridge piers and flooding actions such as hydrodynamic forces from river flow

and scour at underwater foundations.

CRedit authorship contribution statement

José Campos e Matos: Writing – review & editing, Writing – original draft, Supervision, Funding acquisition. **McDonald Hazel:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration. **Marcello Falco:** Methodology, Data curation. **Bartolomeo Panto:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis. **Davide Rapicavoli:** Visualization, Validation, Software, Methodology. **Francesco Cannizzaro:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daniel Oliveira:** Writing – review & editing, Writing – original draft, Supervision. **Enrico Tubaldi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Maurício Bonatte:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ivo Calio:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data will be made available on request.

References

- [1] Orbán Z. UIC Project on assessment, inspection and maintenance of masonry arch railway bridges. 5th Int Conf Arch Bridge 2007:3–12.
- [2] Melbourne C, Wang J, Tomor A, Holm G, Smith M, Bengtsson PE, et al. Masonry arch bridges background document D4.7. Sustain Bridge EU FP 2007;6:277.
- [3] Sarhosis V, De Santis S, de Felice G. A review of experimental investigations and assessment methods for masonry arch bridges. Struct Infrastruct Eng 2016;12: 1439–64. <https://doi.org/10.1080/15732479.2015.1136655>.
- [4] Conde B, Matos JC, Oliveira DV, Riveiro B. Probabilistic-based structural assessment of a historic stone arch bridge. Struct Infrastruct Eng 2020;0:1–13. <https://doi.org/10.1080/15732479.2020.1752261>.
- [5] Moreira VN, Fernandes J, Matos JC, Oliveira DV. Reliability-based assessment of existing masonry arch railway bridges. Constr Build Mater 2016;115:544–54. <https://doi.org/10.1016/j.conbuildmat.2016.04.030>.

- [6] Dhir PK, Losanno D, Tubaldi E, Parisi F. Performance and robustness assessment of roadway masonry arch bridges to scour-induced damage using multiple traffic load models. *Eng Struct* 2025;325:119441. <https://doi.org/10.1016/j.engstruct.2024.119441>.
- [7] Tubaldi E, Macorini L, Izzuddin BA. Three-dimensional mesoscale modelling of multi-span masonry arch bridges subjected to scour. *Eng Struct* 2018;165:486–500. <https://doi.org/10.1016/j.engstruct.2018.03.031>.
- [8] Scozzese F, Ragni L, Tubaldi E, Gara F. Modal properties variation and collapse assessment of masonry arch bridges under scour action. *Eng Struct* 2019;199:109665. <https://doi.org/10.1016/j.engstruct.2019.109665>.
- [9] Page AW. Biaxial Compressive Strength of Brick Masonry. *Proc Inst Civ Eng (Lond) Part 1 - Des Constr* 1981;71:893–906. <https://doi.org/10.1680/iicep.1981.1825>.
- [10] Callaway P, Gilbert M, Smith CC. Influence of backfill on the capacity of masonry arch bridges. *Proc Inst Civ Eng Bridge Eng* 2012;165:147–58. <https://doi.org/10.1680/bren.11.00038>.
- [11] Tubaldi E, Macorini L, Izzuddin BA. Identification of critical mechanical parameters for advanced analysis of masonry arch bridges. *Struct Infrastruct Eng* 2020;16:328–45. <https://doi.org/10.1080/15732479.2019.1655071>.
- [12] Sarhosis V, Liu B, Gilbert M. The 3d response of a large-scale masonry arch bridge - part I: Performance under low and medium loading levels. *Eng Struct* 2024;316:118496. <https://doi.org/10.1016/j.engstruct.2024.118496>.
- [13] Gilbert M, Cole G, Smith C, Melbourne C. Guidance on the assessment of masonry arch bridges (C800). CIRIA; 2022.
- [14] Hughes TG, Blackler MJ. A review of the UK masonry arch assessment methods. *Proc Inst Civ Eng Struct Build* 1997;122:305–15. <https://doi.org/10.1680/istbu.1997.29801>.
- [15] Gilbert M. Limit analysis applied to masonry arch bridges: state-of-the-art and recent developments. 5th Int. Conf. Arch Bridge, Madeira, Portugal: 2007.
- [16] Moreira VN, Matos JC, Oliveira DV. Probabilistic-based assessment of a masonry arch bridge considering inferential procedures. *Eng Struct* 2017;134:61–73. <https://doi.org/10.1016/j.engstruct.2016.11.067>.
- [17] Oliveira DV, Lourenço PB, Lemos C. Geometric issues and ultimate load capacity of masonry arch bridges from the northwest Iberian Peninsula. *Eng Struct* 2010;32:3955–65. <https://doi.org/10.1016/j.engstruct.2010.09.006>.
- [18] Papa T, Grillanda N, Milani G. Three-dimensional adaptive limit analysis of masonry arch bridges interacting with the backfill. *Eng Struct* 2021;248:113189. <https://doi.org/10.1016/j.engstruct.2021.113189>.
- [19] de Felice G. Assessment of the load-carrying capacity of multi-span masonry arch bridges using fibre beam elements. *Eng Struct* 2009;31:1634–47. <https://doi.org/10.1016/j.engstruct.2009.02.022>.
- [20] Reccia E, Milani G, Cecchi A, Tralli A. Full 3D homogenization approach to investigate the behavior of masonry arch bridges: The Venice trans-lagoon railway bridge. *Constr Build Mater* 2014;66:567–86. <https://doi.org/10.1016/j.conbuildmat.2014.05.096>.
- [21] Zizi M, Chisari C, Matteis GDe. Effects of pre-existing damage on vertical load-bearing capacity of masonry arch bridges. *Eng Struct* 2024;300:117205. <https://doi.org/10.1016/j.engstruct.2023.117205>.
- [22] Zhang Y, Tubaldi E, Macorini L, Izzuddin BA. Mesoscale partitioned modelling of masonry bridges allowing for arch-backfill interaction. *Constr Build Mater* 2018. <https://doi.org/10.1016/j.conbuildmat.2018.03.272>.
- [23] Tubaldi E, Minga E, Macorini L, Izzuddin BA. Mesoscale analysis of multi-span masonry arch bridges. *Eng Struct* 2020;225:111137. <https://doi.org/10.1016/j.engstruct.2020.111137>.
- [24] Sarhosis V, Bagi K, Lemos JV, Milani G. Computational Modeling of Masonry Structures Using the Discrete Element Method. IGI Global; 2016.
- [25] Milani G, Lourenço PB. 3D non-linear behavior of masonry arch bridges. *Comput Struct* 2012;110–111:133–50. <https://doi.org/10.1016/j.compstruc.2012.07.008>.
- [26] Pulatsu B, Erdogmus E, Lourenço PB. Simulation of masonry arch bridges using 3D discrete element modeling. In: Aguilar R, Torrealva D, Moreira S, Pando MA, Ramos LF, editors. *Struct. Anal. Hist. Constr.*, 18. Cham: Springer International Publishing; 2019. p. 1949–58. <https://doi.org/10.1007/978-3-319-99441-3>.
- [27] Sarhosis V, Forgács T, Lemos JV. A discrete approach for modelling backfill material in masonry arch bridges. *Comput Struct* 2019;224:106108. <https://doi.org/10.1016/j.compstruc.2019.106108>.
- [28] Pulatsu B, Erdogmus E, Lourenço PB. Comparison of in-plane and out-of-plane failure modes of masonry arch bridges using discontinuum analysis. *Eng Struct* 2019;178:24–36. <https://doi.org/10.1016/j.engstruct.2018.10.016>.
- [29] Calio I, Marletta M, Pantò B. A simplified model for the evaluation of the seismic behaviour of masonry buildings. *Proc 10th Int Conf Civ Struct Environ Eng Comput* 2005. <https://doi.org/10.4203/ccp.81.195>.
- [30] Calio I, Marletta M, Pantò B. A new discrete element model for the evaluation of the seismic behaviour of unreinforced masonry buildings. *Eng Struct* 2012;40:327–38. <https://doi.org/10.1016/j.engstruct.2012.02.039>.
- [31] Pantò B, Cannizzaro F, Calio I, Lourenço PB. Numerical and Experimental Validation of a 3D Macro-Model for the In-Plane and Out-Of-Plane Behavior of Unreinforced Masonry Walls. *Int J Arch Herit* 2017;11:946–64. <https://doi.org/10.1080/15583058.2017.1325539>.
- [32] Cannizzaro F, Pantò B, Caddemi S, Calio I. A discrete macro-element method (DMEM) for the nonlinear structural assessment of masonry arches. *Eng Struct* 2018;168:243–56. <https://doi.org/10.1016/j.engstruct.2018.04.006>.
- [33] Caddemi S, Calio I, Cannizzaro F, D'Urso D, Occhipinti G, Pantò B, et al. A 'Parsimonious' 3d discrete macro-element method for masonry arch bridges. *10thInternational Mason Conf Milan* 2018.
- [34] Caddemi S, Calio I, Cannizzaro F, D'Urso D, Pantò B, Rapicavoli D, et al. 3D discrete macro-modelling approach for masonry arch bridges. *IABSE Symp Guimaraes 2019 Towar a Resilient Built Environ Risk Asset Manag - Rep* 2019:1825–35. <https://doi.org/10.2749/guimaraes.2019.1825>.
- [35] Penazzato L, Oliveira DV, Rapicavoli D, Zampieri P, Lourenço PB, Calio I, et al. Numerical Investigation of a Medieval Masonry Arch Bridge Based on a Discrete Macro-element Modeling Approach. *Proc. 1st Conf. Eur. Assoc. Qual. Control Bridge. Struct.*, 200. Springer International Publishing; 2022. p. 594–603. https://doi.org/10.1007/978-3-030-91877-4_68.
- [36] Rapicavoli D, Cannizzaro F, Caddemi S, Calio I. Discrete macro-element structural assessment of a railway masonry arch bridge subjected to pier settlements. *Life-Cycle Struct Infrastruct Syst* 2023:1590–7. <https://doi.org/10.1201/9781003323020-196>.
- [37] Ahmad SHS, Augustus-Nelson L, Swift GM. Two-dimensional experimental and numerical modelling of a soil-filled masonry arch. *ICE Proc XVI ECSMGE Geotech Eng Infrastruct Dev* 2015:541–6. <https://doi.org/10.1680/ecsmge.60678>.
- [38] Melbourne C, Gilbert M. The behaviour of multiring brickwork arch bridges. *Struct Eng* 1995;73:39–47.
- [39] Melbourne C, Gilbert M, Wagstaff M. The collapse behaviour of multispan brickwork arch bridges. *Struct Eng* 1997;75:297–305.
- [40] Gruppo Sismica. Manual - HiStrA Arches and Vaults and HiStrA Bridges. Catania: 2022.
- [41] Calio I, Cannizzaro F, Marletta M. A discrete element for modeling masonry vaults. *Adv Mater Res* 2010;133–134:447–52. <https://doi.org/10.4028/www.scientific.net/AMR.133-134.447>.
- [42] Vadalà F, Cusmano V, Funari MF, Calio I, Lourenço PB. On the use of a mesoscale masonry pattern representation in discrete macro-element approach. *J Build Eng* 2022;50. <https://doi.org/10.1016/j.jobte.2022.104182>.
- [43] Caddemi S, Calio I, Cannizzaro F, Pantò B, Rapicavoli D. Discrete macroelement modeling. 2019. <https://doi.org/10.1016/B978-0-08-102439-3.00014-2>.
- [44] Lourenço PBP. Recent advances in masonry structures: micromodelling and homogenization. *Multiscale Model Solid Mech Comput Approaches* 2009:280–300.
- [45] Gilbert M, Pritchard T, Smith C. LimitState:RING Manual Version 4, 1. Sheffield, UK: LimitState Ltd; 2025.
- [46] Nela B, Pingaro M, Trovalusci P, Pasca M. Limit Anal multi-ring Mason arches a Parametr Study Eff Frict angle Geom Inter ring Number 2023;2023:1–17. <https://doi.org/10.3389/fbuil.2023.1309696>.
- [47] Gilbert M, Melbourne C. Rigid-block analysis of masonry structures. *Struct Eng* 1994;72.
- [48] Drosopoulos GA, Stavroulakis GE, Massalas CV. Limit analysis of a single span masonry bridge with unilateral frictional contact interfaces 2006;28:1864–73. <https://doi.org/10.1016/j.engstruct.2006.03.016>.
- [49] Malena M, Angelillo M, Fortunato A, de Felice G, Mascolo I. Arch bridges subject to pier settlements: continuous vs. piecewise rigid displacement methods. *Meccanica* 2021;56:2487–505. <https://doi.org/10.1007/s11012-021-01397-1>.
- [50] Zampieri P, Tecchio G. Limit analysis of transverse seismic capacity of multi-span masonry arch bridges 2014. <https://doi.org/10.1007/s10518-014-9664-3>.
- [51] Baggio C, Trovalusci P. Limit analysis for no-tension and frictional three-dimensional discrete systems. *Mech Struct Mach* 1998;26:287–304. <https://doi.org/10.1080/08905459708945496>.